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parameters should be as thorough as practicable.

In table 2 below, various biofouling risk parameters are presented with a description of possible risk impact.

RESOLUTION MEPC.378(80) (adopted on 7 July 2023)
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Table 2: Biofouling risk parameters

Examples of biofouling risk parameters	Description and evaluation guidance
1 Deviation from AFS specifications (e.g. speed, salinity, temperature)	An AFS/AFC can typically work well within a specific range of operating parameters. The relevant parameters and acceptable ranges for each parameter should be described in the manufacturer's specification and included in the BFMP.

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be described in the manufacturer's specification and included in the BFMP.

Specifications typically include operation routes, ship activity level, speed, water salinity and temperature and cleaning requirements. Specifications may va ry depending on the technology of the AFS used.

Ship operations should be in accordance with the recommendations from the AFC manufacturer. Deviation from the specification of the ship's AFC may increase the deterioration of the AFC or reduce its efficacy and change the biofouling risk.

Incidental deviations should be evaluated for potential biofouling impact. Continuous or regular deviations, or deviations not rectified, would lead to an elevated risk profile and more frequent inspection should be part of the contingency plan.

2 Deviation from AFS maintenance/service regime	Regular maintenance and service (e.g. calibration or adjustment of treatment dosages for an MGPS) may be necessary actions fo r proper protection by the

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maintenance/service regime

Regular maintenance and service (e.g. calibration or adjustment of treatment dosages for an MGPS) may be necessary actions fo r proper protection by the
AFS. If the maintenance and service time is exceeded, as specified by the manufacturer, the risk profile is elevated.
For maintenance of AFC, see item 7.

Missing maintenance and/or service should be evaluated as part of the contingency plan for potential biofouling impact.
3 Deviation from regular
proactive cleaning or
necessary reactive cleaning
When proactive cleaning is part of the ship-specific BFMP, deviation from regular use as specified in the BFMP may lead to increased risk of biofouling growth
onto relevant areas. The impact should be evaluated as part of the contingency action plan for potential biofouling impact until the missing proactive cleaning is

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onto relevant areas. The impact should be evaluated as part of the contingency action plan for potential biofouling impact until the missing proactive cleaning is
back in regular operation. Ships should be aware of possible macrofouling accumulation and, if fouling rating is >1, cleaning with capture is the recommended
cleaning action.

If reactive cleaning is not conducted when inspection has determined cleaning is necessary, it will increase the risk of spre ading organisms to new locations.
This risk should be evaluated as part of the contingency plan until the next cleaning event is undertaken.
4 Extended ship idle time Biofouling accumulation starts immediately when a s hip is idle, but the rate depends on AFS type and biofouling pressure (temperature, distance to coast).
To avoid risk of biofouling, the operating profile should only allow short periods in port or at anchorage or at least not exceed the recommendation by the AF S

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To avoid risk of biofouling, the operating profile should only allow short periods in port or at anchorage or at least not exceed the recommendation by the AF S
manufacturer. Acceptable idle time should be specified in the ship's BFMP.

Idle time is often defined in charter party contracts and typically ranges between 18 to 30 days.
If the idle time is longer than specified in the BFMP, the risk profile changes.
If the number of consecutive idle days is still within what is specified as acceptable as per AFS supplier's guarantee and/or idling takes place in an area far from
shore (>200 nm and >200 m depth), the risk may still be considered low.

If the number of consecutive idle days is beyond what is specified as acceptable as per AFS supplier's guarantee, the risk ma y be considered very high if the
ship is subject to biofouling pressure. For these cases, the contingency action plan should include immediate actions before the next voyage.

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ship is subject to biofouling pressure. For these cases, the contingency action plan should include immediate actions

Examples of biofouling risk parameters
Description and evaluation guidance
5 Performance loss as per PMS

Performance monitoring of fuel consumption may give indication on possible biofouling accumulation on the hull. Performance monitoring is mainly for hull monitoring (not niche areas) and may include the following methods:
.1 Sensors and collecting high frequency data.
.2 Semi-automatic or manual calculations using data collected by ship's crew (e.g. noon reports).
.3 Speed trials and comparing the performance data with previous speed trial reports.

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.2 Semi-automatic or manual calculations using data collected by ship's crew (e.g. noon reports).
.3 Speed trials and comparing the performance data with previous speed trial reports.

Note that PMS is often a lagging indicator and may depend on many factors, therefore additional measures may be necessary before it can be used to determine biofouling accumulation.

For some ships, a speed loss between 1% and 3% or increased fuel consumption of 3 -9% may indicate light biofouling while a speed loss >3% or fuel consumption increase by >9% may indicate higher biofouling risk (examples taken from ISO 19030-2:2016).
6 AFS damage Failure caused by mechanical damage to the AFS may result in higher risk of biofouling in the areas affected, if not rectified within reasonable time . Failures and damage should be recorded in the BFRB.

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and damage should be recorded in the BFRB.

As part of the contingency action plan, t he impact from the damage should be evaluated for potential biofouling accumulation and relevant actions should be implemented until a repair is undertaken.
7 Downtime/malfunction of MGPS, proactive cleaning or other AFS
Observed downtime of an MGPS, proactive cleaning or other AFS has a direct impact on risk of biofouling accumulation. The imp act on the area impacted will be affected depending on the duration of malfunction. The impact should be evaluated as part of the contingency action plan for potential biofouling impact until the missing MGPS/proactive cleaning/other AFS is back in operation.

Reduced operation time of proactive cleaning, i.e. longer intervals between cleaning than specified in the BFMP, is

defined a s downtime and may increase

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Reduced operation time of proactive cleaning, i.e. longer intervals between cleaning than specified in the BFMP, is defined a s downtime and may increase
biofouling accumulation particularly in those areas where it is not applied as specified in the BFMP. The impact on the area affected depends on the duration of
malfunction and the trading conditions during that time. The evaluation of impact and potential reactions should be part of the contingency action plan.

If proactive cleaning without capture is irregular, ships should be aware of possible macrofouling accumulation and take actions to avoid spread of macrofouling.
If fouling growth exceeds fouling rating 1, cleaning with capture is recommended.

8 Exceeding expected lifetime
of AFS
Once an AFS has exceeded its lifetime, as specified by the manufacturer, the biofouling risk profile is elevated. Inspection and cleaning should be performed
more often and 1-2 months interval between inspections is recommended.

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more often and 1-2 months interval between inspections is recommended.

Additionally, the efficacy of the AFS may be reduced as it approaches the end of its lifetime. If macrofouling has been remov ed in a previous cleaning event,
the strong forces needed for removing the fouling can have compromised the lifetime of the AFC.

The performance of the AFS, and any necessary change in maintenance or inspection schedule, as given by the AFS manufacturer, should be part of the
contingency action plan specified in the BFMP.

4 Flow chart visualizing biofouling management

An example of a flow chart for visualizing biofouling management risk profile and monitoring of parameters is shown in figure 3.

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4 Flow chart visualizing biofouling management

An example of a flow chart for visualizing biofouling management risk profile and monitoring of parameters is shown in figure 3.

Figure 3: Flow chart visualizing the biofouling management risk profile and monitoring parameters

E tended idle time